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A branch-and-cut algorithm for job-shop scheduling under Time-of-Use pricing and a peak power limit☆

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ABSTRACT

We address the job-shop scheduling problem under Time-of-Use pricing and a peak power limit, with the objective of minimizing total energy costs. To solve this problem, we propose a period-indexed mixed-integer linear programming formulation that reduces the number of variables compared to traditional time-indexed approaches, at the expense of an exponential number of constraints required to enforce the peak power limit. For this reason, we embed the formulation in a branch-and-cut scheme, where peak-power constraints are added only when violated. In addition, the formulation is strengthened with valid inequalities to improve the root relaxation and enumeration tree exploration.

The proposed branch-and-cut algorithm incorporates clique-forbidding cuts, derived from minimal covers of the knapsack polytope associated with the peak-power constraint. We further extend this approach by deriving cuts from arbitrary valid inequalities of the peak-power polytope, and implement a problem-specific cut selection strategy inspired by established metrics in the literature to add the most effective lazy cuts. The resulting framework addresses the structural challenges of energy-aware job-shop scheduling while advancing exact methods in this domain.

1. Introduction

As global energy demands rise, volatile prices and growing concerns about climate change have made energy efficiency a critical focus for industries worldwide. The industrial sector, which accounts for over half of global end-use energy consumption, is expected to see electricity usage grow significantly, from 22% in 2021 to 46% by 2050, due to the electrification of processes (International Energy Agency, 2023). Within this sector, manufacturing systems are major contributors to energy consumption and must adapt to meet increasing demand for goods while managing their energy-intensive nature. Improving energy efficiency in these systems is essential for achieving sustainability goals and reducing environmental impact.

Energy cost minimization offers a strategic approach to enhancing efficiency, particularly through the adoption of Time-of-Use (ToU) pricing. ToU pricing schemes assign varying energy costs to different time periods, creating an incentive for industries to shift production to off-peak times, thereby reducing energy costs, easing grid strain, and lowering greenhouse gas emissions.

The Job-Shop Scheduling Problem (JSSP) is a well-known combinatorial optimization problem, recognized as $\mathcal{NP}$-hard (Garey et al., 1976). It involves sequencing operations on dedicated machines, where each operation must be processed in a predefined order without overlapping. The traditional objective is to minimize the makespan, which represents the completion time of the last operation. However, when considering energy constraints—such as ToU pricing and peak power limits—the problem becomes significantly more complex. Scheduling must not only optimize production but also account for fluctuating energy prices and real-world limitations.

This paper addresses the JSSP under energy considerations, specifically focusing on minimizing total energy costs while adhering to a ToU pricing scheme and a peak power limit. The problem combines the inherent challenges of scheduling with energy optimization. While motivated by industrial contexts, the present contribution is primarily methodological, advancing exact formulations and solution methods for energy-aware scheduling. By focusing on exact methods, we expand the frontier of problems that can be solved to proven optimality, thereby enabling more reliable benchmarking and evaluation of heuristic approaches on a broader set of reference instances. To this end, we propose a new period-indexed Mixed-Integer Linear Programming (MILP) formulation, where variables are indexed on the ToU periods. This yields a model with fewer variables than traditional time-indexed formulations—where each variable indicates whether an

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operation is active at a given discrete time—and avoids relying solely on disjunctive constraints. The peak power limit is enforced through clique-forbidding constraints, derived from the minimal covers of the knapsack set associated with the peak power limit. Since the number of constraints grows exponentially, a branch-and-cut framework is used to dynamically add them as needed during the solution process. We further strengthen the formulation by introducing valid inequalities, while maintaining a compact model size. These enhancements lead to a formulation that outperforms existing benchmarks in the literature. Furthermore, we extend our approach by addressing a Turán-type problem for interval graphs, then generalize the separation scheme with a formula to derive clique-forbidding constraints from any valid inequality of the peak-power constraint polytope. Finally, we implement a cut selection strategy, using problem-specific measures, to prioritize the most effective cuts at each node of the branch-and-cut tree. Together, these elements result in an exact framework that can be extended to other shop-scheduling problems incorporating ToU pricing and resource constraints.

The remainder of the paper is structured as follows: Section 2 provides an overview of related literature. Section 3 introduces key notations and presents the proposed mathematical formulation, while Section 4 introduces and discusses the valid inequalities used to strengthen it. Section 5 presents the algorithm to separate the clique-forbidding inequalities from minimal covers and Section 6 extends it to any valid inequality of the peak power polytope. Section 7 presents the adopted lazy cut selection scheme. Section 8 introduces benchmark sets for the problem, details and analyzes the computational experiments. Finally, Section 9 presents the conclusions of this work and outlines potential directions for future research.

2. Literature review

This paper addresses energy-aware scheduling by minimizing energy cost under ToU pricing and a peak power limit. We first review the literature focusing on similar objectives in different scheduling environments. Next, we discuss selected works centered on energy consumption and, finally, review studies using period-indexed formulations for total energy cost minimization under ToU pricing.

Energy cost minimization under ToU pricing has been explored across various machine scheduling problems. Gaggero et al. (2023) propose a compact MILP with symmetry-breaking properties and a heuristic for a multi-objective problem, addressing both total energy cost (TEC) and makespan on identical parallel machines. Tian and Zheng (2024) develop a branch-and-price (B&P) algorithm for a single-machine batch scheduling problem with TEC minimization. Yuan et al. (2025) study single-machine scheduling with group processing, using MILP formulations and heuristic search paradigms to minimize TEC.

In shop scheduling environments, several works address energy cost minimization under time-of-use pricing using different modeling and solution paradigms. Wang et al. (2018) propose a time-indexed MILP for a two-machine permutation flow shop to minimize TEC while Aghelinejad et al. (2020) study a variant without permutation. Shen et al. (2023) investigate a flexible job-shop problem under time-of-use electricity tariffs, relying on time-indexed formulations and matheuristic approaches to handle the resulting large-scale models. Alternative modeling frameworks have also been explored: Park and Ham (2022) tackle a flexible job-shop problem with machine states using a Constraint Programming (CP) model, minimizing both the makespan and TEC, while Jia et al. (2025) propose a period-indexed (PI) formulation and a matheuristic framework for a two-stage hybrid permutation flow shop, addressing both objectives. Other works address TEC under operational constraints. For example, Masmoudi et al. (2019) introduce disjunctive (D) and time-indexed (TI) formulations, along with a matheuristic, for job-shop scheduling with makespan and peak-power constraints. Bley and Linß (2022) develop a branch-and-bound algorithm (B&B) based on a TI formulation and enhanced by preprocessing and propagation techniques, for job-shop scheduling with release and due dates while accounting for machine states. A comprehensive overview of machine and shop scheduling problems under ToU-based energy tariffs is provided by Catanzaro et al. (2023).

Table 1
Three-field notation for scheduling problems with energy considerations.

Field	Characteristic	Meaning
β field	P^{max}	Peak power limit
	E_k^{max}	Contracted energy consumption limit over a pricing cycle
	on/off	Machine regulation via "on/off" switching mechanism
γ field	TEC	Total energy cost
	TE	Total energy (consumption)

The literature on energy-aware scheduling without ToU pricing is more limited. In the flexible job-shop environment, Meng et al. (2019) propose a MILP based on job sequences minimizing TEC and makespan and accounting for machine states, while Gong et al. (2024) develop a heuristic for the same problem. In machine scheduling, Benedikt et al. (2018) formulate a MILP to minimize total energy consumption, incorporating machine states, release and due dates. Módos et al. (2017) use a B&B and a Logic-Based Benders Decomposition (LBBD) to address a single machine problem with release and due dates, minimizing tardiness under maximum power usage per pricing cycle.

Finally, we highlight works that share methodological similarities with this paper, specifically in their use of period-indexed MILP formulations under ToU pricing. In single machine scheduling, Cheng et al. (2016) compare time-indexed and period-indexed formulations to minimize TEC. For unrelated parallel machines, Che et al. (2017) propose a period-indexed model with valid inequalities and a two-stage heuristic, while Ding et al. (2016) use a column generation heuristic. Aliabad et al. (2020) introduce additional valid inequalities and a relax-and-fix heuristic for the same problem. Wu et al. (2023) present a period-indexed formulation and heuristic for single machine scheduling with release dates. Comparatively, research on shop scheduling with period-indexed formulations is more limited. Ho et al. (2022) propose an LBBD approach for a two-machine permutation flow-shop problem based on Johnson's rule (Johnson, 1954), and Felloussi et al. (2025) propose a period-indexed model enhanced with valid inequalities for job-shop scheduling under TEC minimization.

To formally classify the discussed problems, we use the α, β and γ fields in the conventional three-field notation (Graham et al., 1979) and additional characteristics specific to energy considerations in scheduling problems, listed in Table 1. We adopt the naming conventions of Pinedo (2016) for scheduling problems. For MILP approaches, we specify the nature of the formulation using the previously introduced acronyms D, TI, and PI. The works reviewed are listed in Table 2.

3. Mathematical programming for the energy-efficient job-shop scheduling problem

3.1. Notation and formal problem definition

We consider a set of *machines*, denoted by $\mathcal{M}$, and a set of *jobs*, denoted by $\mathcal{J}$, to be processed over a time horizon C. Each job $j \in \mathcal{J}$ must be executed on a given ordered subset of machines, $\mathcal{M}_j \subseteq \mathcal{M}$, resulting in a series of *operations* that are subject to precedence constraints. Without loss of generality, we assume that every job must be processed on all the machines. Each operation $(j,m) \in \mathcal{O}$, where $\mathcal{O} := \mathcal{J} \times \mathcal{M}$, represents the processing of job j on machine m. We define the precedence relationship $(j,m) \prec (j,m')$ to indicate that operation (j,m) must be performed before operation (j,m'), meaning that in the ordered machine subset $\mathcal{M}_j$, m precedes m'.

The processing time $q_{j,m}$ for an operation (j,m) is known and deterministic. Each machine m has a constant nominal power φ_m, and the energy consumption for processing operation (j,m) is given by $\varphi_m q_{j,m}$.

Table 2
A summary of the cited works on energy-aware scheduling.

Problem class	Reference	Problem	Solution approach
Job-shop scheduling	*This work*	Jm|Pmax|TEC	MILP (PI), B&C
	Felloussi et al. (2025)	Jm||TEC	MILP (PI)
	Bley and Linß (2022)	Jm| on/off, r_j, d_j |TEC	MILP (TI), B&B
	Masmoudi et al. (2019)	Jm|Pmax|TEC	MILP (D,TI), MH
Flexible job-shop scheduling	Gong et al. (2024)	FJm| on/off $\|C_{\max}$,TE	MILP (TI), H
	Shen et al. (2023)	FJm||TEC	MILP (TI), MH
	Park and Ham (2022)	FJm| on/off $\|C_{\max}$,TEC	MILP (TI), CP
	Meng et al. (2019)	FJm| on/off $\|C_{\max}$,TE	MILP (TI)
Flow-shop scheduling	Jia et al. (2025)	HF2|*batch*|$C_{\max}$,TEC	MILP (PI), MH
	Ho et al. (2022)	F2|*prmu*, on/off |TEC	MILP (PI,TI), B&C
	Aghelinejad et al. (2020)	F2| on/off |TEC	MILP (TI)
	Wang et al. (2018)	F2|*prmu*, on/off |TEC	MILP (TI), DP
Parallel machine scheduling	Gaggero et al. (2023)	Pm||$C_{\max}$,TEC	MILP (PI), H
	Aliabad et al. (2020)	Rm||TEC	MILP (PI)
	Benedikt et al. (2018)	Pm| on/off, r_j, d_j |TE	MILP (TI), CP, B&P
	Che et al. (2017)	Rm||TEC	MILP (PI), H
	Ding et al. (2016)	Rm||TEC	MILP (PI), CG-H
Single machine scheduling	Yuan et al. (2025)	1|*group*|TEC	MILP (TI,PI), DP, H
	Tian and Zheng (2024)	1|*batch*|TEC	CG-H
	Wu et al. (2023)	1|r_j|TEC	MILP (PI), H
	Módos et al. (2017)	1|$E_k^{\max}, r_j, d_j$| $\sum_j T_j$	B&B, B&C, H
	Cheng et al. (2016)	1|*batch*|TEC	MILP (PI)

CG-H: column generation heuristic
MH: matheuristic
H: problem specific heuristic
DP: dynamic programming
B&B: branch and bound
B&C: branch and cut
B&P: branch and price
CP: constraint programming
TI: time-indexed
D: disjunctive
PI: period-indexed

The energy usage is constrained by a peak power limit $\overline{\varphi}$ and subject to a ToU pricing scheme dividing the time horizon C into a set $\mathcal{P}$ of periods. Each period $p \in \mathcal{P}$ has a duration l^p and a price per energy unit κ^p. Let t^p denote the start time of period p, with $t^1 = 0$ and $t^{p+1} - t^p = l^p$ for all $p \in 1, \dots, |\mathcal{P}|$. The end of the last period is given by $t^{|\mathcal{P}|+1} = C$.

The goal is to determine a schedule within the time horizon C that minimizes the total energy cost, as determined by the ToU pricing profile, while ensuring that all operations comply with non-preemption, non-overlap, precedence, and the peak power limit constraints.

3.2. Decision variables and parameters

Let symbols $j \in \mathcal{J}$, $m \in \mathcal{M}$ and $p \in \mathcal{P}$ respectively denote jobs, machines and ToU periods. We write $\mathbb{N}^*$ for the set of positive integers. The decision variables are (i) binary *processing-status* variables $x^p_{j,m}$, equal to 1 if operation (j,m) is processed during p, and 0 otherwise, (ii) nonnegative *processing-duration* variables $d^p_{j,m}$, equal to the duration of processing of (j,m) during p, (iii) non-negative *start-date* variables $s_{j,m}$, denoting the start date of (j,m), (iv) binary *operation-precedence* variables $u^{j',m'}_{j,m}$, equal to 1 if (j,m) completes before (j',m') starts, and 0 otherwise. For brevity, we define auxiliary expressions $c_{j,m} := s_{j,m} + q_{j,m}$, the completion date of (j,m), and $z^{j',m'}_{j,m} := 1 - u^{j',m'}_{j,m} - u^{j,m}_{j',m'}$, the overlap indicator for pairs with $j \neq j'$ and $m \neq m'$ (operations sharing the job or machine cannot overlap); we take $j < j'$ to avoid duplicates.

Let $\underline{s}_{j,m}$ and $\overline{s}_{j,m}$ (resp. $\underline{c}_{j,m}$ and $\overline{c}_{j,m}$) denote the earliest and latest starting (resp. completion) dates for each operation (j,m), determined through precedence relationships. We use these values to compute an upper bound $\overline{d}^p_{j,m}$ on the processing duration of operation (j,m) within period p. It is defined as the length of the intersection between the feasible execution window $[\underline{s}_{j,m}, \overline{c}_{j,m}]$ and the period interval $[t^p, t^{p+1}]$, and can be expressed as:

$$\overline{d}^p_{j,m} = \begin{cases} 0 & \text{if } t^{p+1} \leqslant \underline{s}_{j,m} \text{ or } \overline{c}_{j,m} \leqslant t^p, \\ t^{p+1} - \underline{s}_{j,m} & \text{if } t^p \leqslant \underline{s}_{j,m} \leqslant t^{p+1}, \\ \overline{c}_{j,m} - t^p & \text{if } t^p \leqslant \overline{c}_{j,m} \leqslant t^{p+1}, \\ q_{j,m} & \text{if } t^p \leqslant \underline{s}_{j,m} \text{ and } \overline{c}_{j,m} \leqslant t^{p+1}, \\ l^p & \text{otherwise.} \end{cases}$$

Let $\mathcal{V} := \mathcal{O}$ and let $\mathcal{E}$ denote the set of all admissible overlap edges $e = ((j,m),(j',m'))$ with $j < j'$ and $m \neq m'$. For each $e \in \mathcal{E}$, we write z^e for the corresponding overlap indicator $z^{j',m'}_{j,m}$. We consider the *overlap graph* $\mathcal{G}(z) = (\mathcal{V}, \mathcal{E}(z))$ induced by z, where $\mathcal{E}(z) := \{e \in \mathcal{E} : z^{j',m'}_{j,m} = 1\}$. Each vertex $v \in \mathcal{V}$ is assigned a weight φ_v, equal to the power demand of the machine on which the corresponding operation executes. A *clique* $K \subset \mathcal{V}$ of $(\mathcal{V}, \mathcal{E})$ is a set of pairwise adjacent vertices; we denote its edge set by $E(K)$ and its weight by $\varphi(K) := \sum_{v \in K} \varphi_v$, equal to the total power demand of the operations in K.

3.3. Mathematical model

The Jm|Pmax|TEC can be modeled as the following program:

$$\min \quad \sum_{p \in \mathcal{P}} \kappa^p \sum_{m \in \mathcal{M}} \varphi_m \sum_{j \in \mathcal{J}} d^p_{j,m} \tag{1}$$

$$\text{s.t.} \quad \sum_{p \in \mathcal{P}} d^p_{j,m} = q_{j,m}, \qquad \forall (j,m) \in \mathcal{O}, \tag{2}$$

$$d^p_{j,m} \leqslant \overline{d}^p_{j,m} x^p_{j,m}, \qquad \forall (j,m) \in \mathcal{O}, \forall p \in \mathcal{P}, \tag{3}$$

$$d^p_{j,m} \leqslant t^{p+1} - s_{j,m} + \overline{s}_{j,m}(1 - x^p_{j,m}), \qquad \forall (j,m) \in \mathcal{O}, \forall p \in \mathcal{P}, \tag{4}$$

$$d^p_{j,m} \leqslant c_{j,m} - t^p x^p_{j,m}, \qquad \forall (j,m) \in \mathcal{O}, \forall p \in \mathcal{P}, \tag{5}$$

$$c_{j,m} \leqslant s_{j,m'}, \qquad \forall j \in \mathcal{J}, \forall m, m' \in \mathcal{M}_j : (j,m) \prec (j,m'), \tag{6}$$

$$c_{j,m} - s_{j',m'} \leqslant (\overline{c}_{j,m} - \underline{s}_{j',m'})(1 - u^{j',m'}_{j,m}), \qquad \forall (j,m),(j',m') \in \mathcal{O} : j \neq j', \tag{7}$$

$$u^{j',m}_{j,m} + u^{j,m}_{j',m} = 1, \qquad \forall m \in \mathcal{M}, \forall j, j' \in \mathcal{J} : j < j', \tag{8}$$

$$\sum_{e \in E(K)} z^e \leqslant |E(K)| - 1, \qquad \forall K \text{ clique in } (\mathcal{V}, \mathcal{E}) : \varphi(K) > \overline{\varphi}, \tag{9}$$

$$d^p_{j,m} \geqslant 0, x^p_{j,m} \in \{0,1\}, \qquad \forall (j,m) \in \mathcal{O}, \forall p \in \mathcal{P}, \tag{10}$$

$$\underline{s}_{j,m} \leqslant s_{j,m} \leqslant \overline{s}_{j,m}, \qquad \forall (j,m) \in \mathcal{O}, \tag{11}$$

$$u^{j',m'}_{j,m} \in \{0,1\}, \qquad \forall (j,m),(j',m') \in \mathcal{O} : j \neq j'. \tag{12}$$

The objective function (1) minimizes the total operational cost of a schedule. Constraints (2) ensure that the operations are fully executed over the horizon. Constraints (3) link processing durations to the binary processing-status variables, and constraints (4)–(5) relate these durations to the start and completion dates. Constraint (6) enforces the precedence relationships between operations of the same job. Further, the operation precedence variable is defined in (7), and constraint (8) ensures that any two operations requiring the same machine do not overlap.

The peak-power constraint (9) forbids any clique K whose weight exceeds the peak power limit. As we show below, it suffices to consider the minimal covers of the knapsack set associated with this constraint. Let $n := |\mathcal{M}|$. The *0,1 knapsack set* associated with the peak-power constraint is defined as $\mathbf{KP}^{\varphi,\overline{\varphi}} := \left\{x \in \{0,1\}^n : \sum_{j=1}^{n} \varphi_j x_j \leqslant \overline{\varphi}\right\}$. The knapsack polytope of $\mathbf{KP}^{\varphi,\overline{\varphi}}$ will be denoted as $\mathbf{P}^{\varphi,\overline{\varphi}} := \text{conv}\{\mathbf{KP}^{\varphi,\overline{\varphi}}\}$, where $\text{conv}\{\cdot\}$ is the convex hull operator. A subset $C \subseteq [n] := \{1,\dots,n\}$ of indices is called a *cover* of $\mathbf{KP}^{\varphi,\overline{\varphi}}$ if $\sum_{i\in C} \varphi_i > \overline{\varphi}$, and it is a *minimal cover* if it does not contain a proper subset that is a cover, i.e., $\sum_{i\in C\setminus\{j\}} \varphi_i \leqslant \overline{\varphi}$ for every $j \in C$. By definition of minimal covers, any proper subset of vertices of K has a weight that does not exceed the limit. To formalize, we say that a clique K is $\overline{\varphi}$*-minimal* if $\varphi(K) > \overline{\varphi}$ and $\varphi(\mathcal{H}) \leqslant \overline{\varphi}$ for every proper subset of vertices $\mathcal{H} \subsetneq K$. This yields the following peak-power constraint, which replaces (9):

$$\sum_{e\in E(K)} z^e \leqslant |E(K)| - 1, \quad \forall K \text{ clique of } (\mathcal{V},\mathcal{E}) \text{ that is } \overline{\varphi}\text{-minimal}, \tag{9.a}$$

where $|E(K)| = \binom{|K|}{2}$. The resulting mixed-integer program (1)–(8), (9.a)–(12), denoted in the following as (MIP), solves the Jm|P_{max}|TEC exactly.

Example 1. We provide a brief numerical example to show how the processing duration $d^p_{j,m}$ and processing status $x^p_{j,m}$ variables relate to the date variables $s_{j,m}$. Consider a schedule with $|\mathcal{P}| = 6$, period bounds $t = \{0,1,4,6,8,11,12\}$, and an operation of duration $q = 7$ that starts at $s = 2$ and completes at $c = 9$. The resulting processing-status vector is $x = \{0,1,1,1,1,0\}$.

We have, in accordance with the introduced constraints and as shown in Fig. 1:

- $d^1 = d^6 = 0$ due to (3),
- for $p = 2$, among inequalities (4)–(5), the most constraining is the first one, yielding $d^2 \leqslant 2$,
- for $p = 3$ and $p = 4$, constraints (3) enforce that $d^3 \leqslant l^3 = 2$ and $d^4 \leqslant l^4 = 2$,
- for $p = 5$, the most constraining inequality is the second one of (4)–(5), yielding $d^5 \leqslant 1$,
- these inequalities alongside the partition equality (2) result in $d = \{0,2,2,2,1,0\}$.

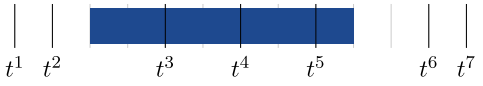


Fig. 1. A one operation example.

Example 2. Regarding the peak-power constraint, consider the following example with five operations (a,b,c,d,e) from distinct jobs, each executed on a distinct machine, and let $n := |\mathcal{J}| = |\mathcal{M}| = 5$. The machine power vector φ is $[1,3,3,4,5]$, and the peak power limit $\overline{\varphi}$ is 15. Then, $C := \{1,2,3,4,5\}$ is a minimal cover since $\sum_{m\in C} \varphi_m > \overline{\varphi}$ and $\sum_{m\in C'} \varphi_m \leqslant \overline{\varphi}$ for every proper subset $C' \subsetneq C$ Fig. 2 represents the corresponding processing intervals and the induced overlap graph. The corresponding clique-forbidding inequality is $z^b_a + z^c_a + z^d_a + z^e_a + z^c_b + z^d_b + z^e_b + z^d_c + z^e_c + z^e_d \leqslant 9$. The left case forms a clique on $\{a,b,c,d,e\}$, which exceeds the peak limit ($16 > \overline{\varphi}$). Therefore, at least one pair of operations must be non-overlapping, i.e., at least one edge must be missing, as enforced by the inequality (right case).

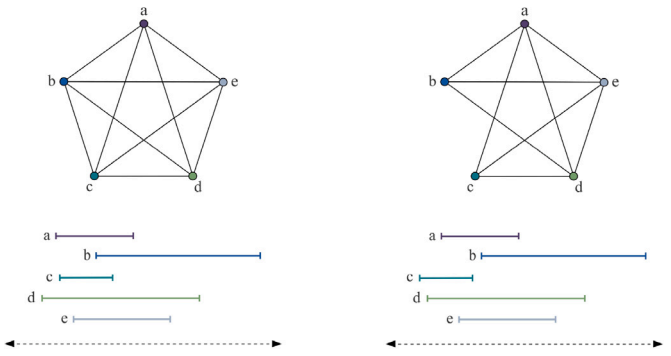


Fig. 2. A set of overlapping operations exceeding (left) and respecting (right) the peak-power constraint.

4. Valid inequalities

It is well-documented that formulations relying on date variables and disjunction variables, and/or relying on big-M constraints, often yield weak LP relaxations (see, e.g., Applegate & Cook, 1991 and Conforti et al., 2014). In what follows, we introduce valid inequalities to strengthen the formulation, addressing both the scheduling and capacity aspects of the problem. Additional details on these inequalities are provided in Appendix 1.

4.1. Scheduling-related valid inequalities

Proposition 1. *The following inequalities are valid for (MIP):*

$$\sum_{j\in\mathcal{J}} d^p_{j,m} \leqslant l^p X^p_m, \qquad \forall m \in \mathcal{M}, \forall p \in \mathcal{P}, \tag{13}$$

$$\sum_{m\in\mathcal{M}} d^p_{j,m} \leqslant l^p, \qquad \forall j \in \mathcal{J}, \forall p \in \mathcal{P}, \tag{14}$$

$$y^p_{j,m} \geqslant x^p_{j,m} + x^{p+1}_{j,m} - 1, \qquad \forall (j,m) \in \mathcal{O}, \forall p \in \{1,\dots,|\mathcal{P}|-1\}, \tag{15}$$

$$\sum_{j\in\mathcal{J}} y^p_{j,m} \leqslant X^p_m, \qquad \forall m \in \mathcal{M}, \forall p \in \{1,\dots,|\mathcal{P}|-1\}, \tag{16}$$

$$\sum_{j\in\mathcal{J}} y^p_{j,m} \leqslant X^{p+1}_m, \qquad \forall m \in \mathcal{M}, \forall p \in \{1,\dots,|\mathcal{P}|-1\}, \tag{17}$$

$$\lceil \frac{q_{j,m}}{\max_{p\in\mathcal{P}} l^p} \rceil \leqslant \sum_{p\in\mathcal{P}} x^p_{j,m} \leqslant \lfloor \frac{q_{j,m}}{\min_{p\in\mathcal{P}} l^p} \rfloor + 2, \qquad \forall (j,m) \in \mathcal{O}. \tag{18}$$

Inequalities (13)–(14) follow directly from the definition of the processing duration variables d. Similar inequalities can be found in Felloussi et al. (2025). We define the binary variable $X^p_m \in \{0,1\}$ via (13), indicating whether machine m is active during period p; this variable also appears in inequalities (16) and (17). Inequality (15) introduces the variable $y^p_{j,m}$, which indicates whether operation (j,m) spans periods p and $p+1$. A similar y-based formulation was first used in Che et al. (2017), while inequalities (18) were introduced in Aliabad et al. (2020).

Proposition 2. *The following inequalities are valid for (MIP):*

$$d^p_{j,m} \geqslant l^p (x^{p_1}_{j,m} + x^{p_2}_{j,m} - 1), \quad \forall (j,m) \in \mathcal{O}, \forall p_1, p, p_2 \in \mathcal{P} : p_1 < p < p_2. \tag{19}$$

Proposition 3. *The following inequalities are valid for (MIP):*

$$\sum_{p'=1}^{p-1} d^{p'}_{j,m'} \leqslant \underline{M}^p_{j,m'} (1 - x^p_{j,m}), \qquad \forall j \in \mathcal{J}, \forall m, m' \in \mathcal{M}_j : (j,m) \prec (j,m'), \forall p \in \mathcal{P}, \tag{20}$$

$$\sum_{p'=p+1}^{|\mathcal{P}|} d^{p'}_{j,m} \leqslant \overline{M}^p_{j,m} (1 - x^p_{j,m'}), \qquad \forall j \in \mathcal{J}, \forall m, m' \in \mathcal{M}_j : (j,m) \prec (j,m'), \forall p \in \mathcal{P}, \tag{21}$$

$$\sum_{p'=1}^{p-1} d_{j,m}^{p'} \leqslant \underline{M}_{j,m}^{p}(1 - x_{j,m}^{p} + x_{j,m}^{p-1}), \quad \forall (j,m) \in \mathcal{O}, \forall p \in \{2,\dots,|\mathcal{P}|\}, \tag{22}$$

$$\sum_{p'=p}^{|\mathcal{P}|} d_{j,m}^{p'} \leqslant \overline{M}_{j,m}^{p}(1 + x_{j,m}^{p} - x_{j,m}^{p-1}), \quad \forall (j,m) \in \mathcal{O}, \forall p \in \{2,\dots,|\mathcal{P}|\}, \tag{23}$$

where $\underline{M}_{j,m}^{p} := \min\left(q_{j,m}, \sum_{p'=1}^{p-1} l^{p'}\right)$, *and* $\overline{M}_{j,m}^{p} := \min\left(q_{j,m}, \sum_{p'=p}^{|\mathcal{P}|} l^{p'}\right)$. *Moreover, when processing times and ToU period durations are integral, the processing duration variables take integer values in optimal solutions (see Lemma 1 in Appendix 1), and the following inequality is valid and strengthens (20)–(23):*

$$x_{j,m}^{p} \leqslant d_{j,m}^{p}, \qquad \forall (j,m) \in \mathcal{O},\ \forall p \in \mathcal{P}. \tag{24}$$

Inequalities (20)–(23) are implied bound cuts. They can be introduced in disaggregated or aggregated form; the disaggregated form is stronger but typically slows down LP solves (Achterberg, 2007). In this study, we use the aggregated form.

4.2. Capacity-related valid inequalities

In the following, we use the expression $D_m^p := \sum_{j\in\mathcal{J}} d_{j,m}^p$ for all machines m and periods p to simplify notation. $\mathbf{P}^{\varphi,\overline{\varphi}}$ denotes the convex hull of the 0,1 knapsack set associated with the peak-power capacity constraint. For $a \in \mathbb{R}^{|\mathcal{M}|}$ and $b \in \mathbb{R}$, we write $a^{\intercal}\mathbf{x} \leqslant b$ for a valid inequality of $\mathbf{P}^{\varphi,\overline{\varphi}}$.

Proposition 4. *The following inequalities are valid for (MIP):*

$$\sum_{m\in\mathcal{M}} a_m D_m^p \leqslant b \times l^p, \qquad \forall p \in \mathcal{P}, \tag{25}$$

$$\sum_{m\in\mathcal{M}} a_m \sum_{j\in\mathcal{J}} y_{j,m}^p \leqslant b, \qquad \forall p \in \mathcal{P}. \tag{26}$$

Inequalities (25) also apply with $a_m = \varphi_m$ and $b = \overline{\varphi}$, resulting in the constraint $\sum_{m\in\mathcal{M}} \varphi_m D_m^p \leqslant \overline{\varphi}\, l^p$ for each period $p \in \mathcal{P}$. Define $\overline{E}_m^p := \varphi_m \frac{D_m^p}{l^p}$. Then $\sum_{m\in\mathcal{M}} \overline{E}_m^p \leqslant \overline{\varphi}$ and, by (13), $\overline{E}_m^p \leqslant \varphi_m X_m^p$ for all $m \in \mathcal{M}$. This identifies, for each $p \in \mathcal{P}$, a fixed-charge polytope (Padberg et al., 1985):

$$\mathbf{P}_{\text{f-c}}^p = \text{conv}\left\{ (\overline{E}^p, X^p) \in \mathbb{R}_+^{|\mathcal{M}|} \times \{0,1\}^{|\mathcal{M}|} : \sum_{m\in\mathcal{M}} \overline{E}_m^p \leqslant \overline{\varphi},\ \overline{E}_m^p \leqslant \varphi_m X_m^p\ \forall m \in \mathcal{M} \right\}. \tag{27}$$

A *flow cover* is a set $F \subseteq \mathcal{M}$ such that $\sum_{m\in F} \varphi_m > \overline{\varphi}$. For a flow cover F, let $\lambda := \sum_{m\in F} \varphi_m - \overline{\varphi}$ and $F^+ := \{m \in F :\ \varphi_m > \lambda\}$. By Padberg et al. (1985), these sets yield valid flow cover inequalities for $\mathbf{P}_{\text{f-c}}^p$. We use their lifted form proposed by Gu et al. (1999), who also provide closed-form expressions for the lifting coefficients (α_m, β_m).

Proposition 5. *For any flow cover F and any $p \in \mathcal{P}$, the inequality*

$$\sum_{m\in F} \overline{E}_m^p + \sum_{m\in F^+} (\varphi_m - \lambda)^+(1 - X_m^p) + \sum_{m\in\mathcal{M}\setminus F} \left(\alpha_m \overline{E}_m^p + \beta_m X_m^p\right) \leqslant \overline{\varphi} \tag{28}$$

is a lifted flow cover inequality and is valid for (MIP).

5. A branch-and-cut algorithm based on minimal covers

In this section, we propose a branch-and-cut procedure to solve (MIP). Since the family of clique-forbidding constraints (9.a) enforcing the peak power limit is exponential in size, solving MIP directly is computationally impractical. While the number of minimal covers is moderate, for each minimal cover $C \in \mathcal{C}$, every injective assignment of $|C|$ jobs to the machines in C defines a distinct clique pattern that must be forbidden. The number of such assignments is the number of $|C|$-permutations of $\mathcal{J}$, and thus the number of resulting constraints over all covers is $\sum_{C\in\mathcal{C}} \frac{|\mathcal{J}|!}{(|\mathcal{J}|-|C|)!}$.

To cope with this exponential number of constraints, we embed the formulation in a branch-and-cut framework. The procedure runs a branch-and-bound algorithm on (MIP) without the clique-forbidding constraints, and whenever an integer solution is encountered at a node of the search tree, a separation procedure is applied to check feasibility with respect to the peak power limit. If violated constraints are found, they are added at the current node and the solution is rejected; otherwise, the solution is accepted as feasible and possibly as the new incumbent. From a correctness standpoint, the addition of any violated clique-forbidding inequality corresponding to a minimal cover is sufficient to exclude the current infeasible solution.

Let an *event* refer to the start or completion of any operation. Consequently, there are at most $2|\mathcal{O}|$ events in any schedule, and the set of active operations (hence the total power demand) is constant between two consecutive operations. The separation procedure performs a window scan over the scheduling horizon, where each window is defined by two consecutive events. Over the interior of a window, the set of active operations is constant; we therefore evaluate the power demand at a representative time point (we use the midpoint for convenience) and check whether it exceeds the peak power limit. If it does, we evaluate all minimal covers for violation and add the corresponding clique-forbidding constraints associated with the operations active at that midpoint; Algorithm 1 summarizes the separation procedure, which runs in time polynomial in the number of operations. This approach is computationally reasonable because the number of minimal covers depends primarily on the number of machines and, to a lesser extent, on the peak power limit. In practice, the number of machines—and thus the size of the associated knapsack polytope—remains relatively small compared with typical knapsack instances, even for large scheduling problems. As there are at most $2|\mathcal{O}|$ events, the window scan over the scheduling horizon runs in linear time.

Separation is applied only to integer solutions. Indeed, the separation procedure relies on well-defined operation start and completion times, and on a unique set of active operations between consecutive events, properties that only hold at integer solutions due to the big-M timing constraints. The resulting clique-forbidding constraints are no-good cuts, sometimes also referred to as combinatorial Benders cuts (Codato & Fischetti, 2006). As such, they eliminate a single infeasible integral point and are weak in the polyhedral sense (Subramanian et al., 2011).

Algorithm 1 Pseudocode of the separation algorithm to cut infeasible integer solutions.

Require: s, c: starting and completion dates
Require: $\mathcal{C}$: set of minimal covers
E ← SortAndUniqueEvents(s, c)
$\mathrm{E}^{\frac{1}{2}}$ ← EventMidPoints(E)
cutPool ← ∅
for each $e \in \mathrm{E}^{\frac{1}{2}}$, **do**
  for each $C \in \mathcal{C}$ **do**
    if $\exists\ m \in C$: m inactive at e **then**
      continue ▷ *C is minimal for inclusion*
    end if
    $\mathcal{O}_{\text{infeas}} \leftarrow \emptyset$
    for each $m \in C$ **do**
      for each $j \in \mathcal{J}$ **do**
        if j is active on m at e **then**
          add (j,m) to $\mathcal{O}_{\text{infeas}}$
        end if
      end for
    end for
    add PowerLimCut($\mathcal{O}_{\text{infeas}}$) to cutPool
  end for
end for
return cutPool

The operator SortAndUniqueEvents constructs a list of events, removes duplicate start and completion dates, and sorts them in non-decreasing order. The operator EventMidPoints returns the set

of event midpoints, and for a set of active operations that result in infeasibility, `PowerLimCut` returns the associated peak power limit cut, i.e., a clique-forbidding inequality.

6. An extension to any valid inequality of the peak-power constraint polytope

So far, we have only separated clique-forbidding inequalities derived from minimal covers. Since these inequalities are model-defining, it suffices to ensure that none of them is violated to obtain feasibility. However, there are valid inequalities that are stronger for the peak-power constraint polytope $\text{conv}\{\mathbf{KP}^{\varphi,\overline{\varphi}}\}$, namely the facet-inducing inequalities describing it that may not be minimal covers. In this section, we show how to obtain cuts from these inequalities, in a form analogous to (9.a). Our construction builds on the extremal number (or Turán number) of the overlap graph induced by the variables z, to derive valid inequalities from extended covers. We then provide a formula that generalizes the resulting inequality to from any valid inequality of $\text{conv}\{\mathbf{KP}^{\varphi,\overline{\varphi}}\}$.

6.1. Clique-forbidding cuts from extended covers

Consider the 0–1 knapsack set $\mathbf{KP}^{\varphi,\overline{\varphi}}$ defined previously, and let $C \subseteq \mathcal{M}$ be a minimal cover of $\mathbf{KP}^{\varphi,\overline{\varphi}}$. We assume without loss of generality that $C' := \{m \in \mathcal{M} \setminus C \mid \varphi_m \geqslant \varphi_{m'}, \forall m' \in C\}$ is non-empty. An *extended cover* of C is any set $C'' := C \cup C'$. Let $n, r \in \mathbb{N}^*$ be such that $n = |C''|$ and $r = |C| - 1$ (so $r < n$). We consider an integer infeasible candidate solution and denote by $G := \mathcal{G}(z)$ the overlap graph induced by the values of the overlap variables z. Let K_n be an n-vertex clique of G. By definition, we can associate to an extended cover C'' an inequality specifying that among n available machines, at most r can execute simultaneously. Equivalently, in G, the subgraph induced by the operations assigned to machines in C'' must be K_{r+1}-free, i.e., it cannot contain a clique of size $r+1$. One approach to enforce this involves studying the (extremal) graph with maximum number of edges while forbidding K_{r+1}. This is a Turán-type problem (see Zhao, 2023 and references therein). Its solution, the Turán number, provides an upper bound on the number of edges in such graphs given by $(1-\frac{1}{r})\frac{n^2}{2}$ (Turán, 1941). This value can be used as a right-hand side in clique-forbidding inequalities analogous to (9.a).

We show that a tighter bound can be established when restricting to *interval graphs*, i.e., graphs whose vertices can be associated with intervals on the real line, with an edge between two vertices if and only if the corresponding intervals intersect. This restriction is natural here because, at integer solutions, each operation defines an execution interval on the timeline, and the resulting overlap graph is precisely the intersection graph of these intervals. In the following, we refer to the K_{r+1}-free graph that maximizes the number of edges as the *r-extremal* graph.

Lemma 1. *Let $n \in \mathbb{N}^*$ and let $I_i := [a_i, b_i]$, $i \in [n]$, be a set of intervals. Without loss of generality, assume that the intervals are indexed in non-decreasing order of their right endpoints, i.e., $b_i \leqslant b_{i+1}$ for all $i \leqslant n-1$. Then, for any $i < j-1$, if $I_i \cap I_j \neq \emptyset$, we have $I_j \cap I_k \neq \emptyset$ for all $i < k < j$.*

Proof. Let $i, j \in [n]$ with $i < j-1$, and assume $I_i \cap I_j \neq \emptyset$. Then $a_j \leqslant b_i$ by interval intersection. For any k such that $i < k < j$, the ordering implies $b_i \leqslant b_k \leqslant b_j$, hence $a_j \leqslant b_k$. Moreover, $a_k \leqslant b_k \leqslant b_j$, so $a_k \leqslant b_j$. Therefore, $a_j \leqslant b_k$ and $a_k \leqslant b_j$, which implies $I_j \cap I_k \neq \emptyset$. □

Theorem 2. *The number of edges in an n-vertex r-extremal interval graph is*

$$\binom{n}{2} - \binom{n-r+1}{2}. \tag{29}$$

Proof. Let $n \in \mathbb{N}^*$ and let $I_i := [a_i, b_i]$, $i \in [n]$, be a set of intervals indexed in non-decreasing order of their right endpoints. Let G denote the associated interval graph on n vertices, and let $\overline{G}$ denote its complement. We assume that G is K_{r+1}-free.

For an index $i \in [n-r]$, we argue by contradiction and assume that I_i intersects at least r intervals among $I_{i+1}, \dots, I_n$. Let $j > i$ be the largest index such that $I_i \cap I_j \neq \emptyset$. Then there exist at least $r-1$ indices k with $i < k < j$ such that $I_i \cap I_k \neq \emptyset$. Since $i < j-1$ and $I_i \cap I_j \neq \emptyset$, Lemma 1 implies that $I_j \cap I_k \neq \emptyset$ for all $i < k < j$. Moreover, for any $\ell > i$ such that $I_i \cap I_\ell \neq \emptyset$, interval intersection and the ordering yield $a_\ell \leqslant b_i \leqslant b_\ell$, and hence $b_i \in I_\ell$. It follows that these $r+1$ intervals are pairwise intersecting, forming a K_{r+1}, which contradicts that G is K_{r+1}-free. Therefore, I_i intersects at most $r-1$ intervals among the later intervals $I_{i+1}, \dots, I_n$.

Equivalently, I_i is disjoint from at least $(n-i)-(r-1)$ later intervals, and each such disjoint pair corresponds to an edge of $\overline{G}$. For $i \leqslant n-r$, we have $(n-i)-(r-1) \geqslant 1$, so I_i forces at least one edge in $\overline{G}$. Summing over $i = 1, \dots, n-r$, we obtain

$$|E(\overline{G})| \geqslant \sum_{i=1}^{n-r} \big((n-i)-(r-1)\big) = \binom{n-r+1}{2}.$$

Since $|E(G)| + |E(\overline{G})| = \binom{n}{2}$, it follows that

$$|E(G)| \leqslant \binom{n}{2} - \binom{n-r+1}{2}.$$

We prove this bound is tight by an explicit construction. Consider r intervals with a common intersection point, forming a K_r, and add $n-r$ intervals that are pairwise disjoint, each intersecting exactly $r-1$ of the r intervals. The resulting interval graph is K_{r+1}-free and has exactly $\binom{n}{2} - \binom{n-r+1}{2}$ edges. By definition, an r-extremal interval graph therefore has $\binom{n}{2} - \binom{n-r+1}{2}$ edges. □

Note 1. A similar result has already been proven by Abbott and Katchalski (1979) as later clarified by Eckhoff (1993). Their formulation of the result is $\binom{r}{2} + (n-r)(r-1)$, which stems from a different proof approach. It is straightforward to verify the equivalence of the two expressions. Our formulation may offer additional insights into the structure of extremal interval graphs.

Note 2. The result of Lemma 1 can be leveraged to propose a MIP formulation that derives the formula computationally. Further details are provided in Appendix 2.

Example 3. Consider a capacity constraint where $\overline{\varphi} = 10$ and $\varphi = [1, 3, 3, 4, 5]$. For $\text{conv}\{\mathbf{KP}^{\varphi,\overline{\varphi}}\}$, the inequality $x_3 + x_4 + x_5 \leqslant 2$ follows from the minimal cover $\{3, 4, 5\}$ ($r = 2$ and $n = 3$), while the inequality $x_1 + x_2 + x_3 + x_4 + x_5 \leqslant 3$ is obtained from extending the minimal cover $\{2, 3, 4, 5\}$ with the singleton $\{1\}$ ($r = 3$ and $n = 5$). These inequalities yield, respectively, the following clique-forbidding inequalities on the overlap variables:

$$z_3^4 + z_3^5 + z_4^5 \leqslant 2,$$
$$z_1^2 + z_1^3 + z_1^4 + z_1^5 + z_2^3 + z_2^4 + z_2^5 + z_3^4 + z_3^5 + z_4^5 \leqslant 7.$$

6.2. Generalization to any valid inequality of the peak-power constraint polytope

Let $n \in \mathbb{N}^*$, $a = (a_1, \dots, a_n) \in \mathbb{R}_+^n$ and $b \in \mathbb{R}_+$ such that $a^\intercal \mathbf{x} \leqslant b$ is valid for $\text{conv}\{\mathbf{KP}^{\varphi,\overline{\varphi}}\}$. The case $a_i \in \{0, 1\}$ yields cover inequalities, which we treated in the previous section. We now derive clique-forbidding inequalities for general coefficient vectors a. Our approach relies on an interval-replication argument: if a vertex v has an integer coefficient $a_v \geqslant 2$, it can be interpreted as a_v identical copies of the same interval, which therefore overlap pairwise. This transformation reduces the weighted setting to an equivalent unit-weight interval graph, allowing

us to apply the same clique-forbidding construction as for minimal and extended covers.

Given an n-vertex graph $\mathcal{G}_a = (\mathcal{E}_a, \mathcal{V}_a)$ with vertices of weights $a = (a_1, \dots, a_n)$, we seek to derive a valid inequality on the *edges* (through variables z), such that cliques of *node*-weight greater than or equal to b are forbidden. This corresponds to finding a coefficient vector α and a right-hand-side β, such that $\sum_{(u,v)\in\mathcal{E}^a} \alpha_{u,v} z_u^v \leqslant \beta$ is valid with respect to the capacity constraint for (MIP).

Let $\mathcal{I} := \{v \in \mathcal{V}_a, a_v \geqslant 2\}$. For each $v \in \mathcal{I}$, we replace v by a_v identical copies, all associated with the same interval as v. Let $\mathcal{V}_a^1$ denote the resulting multiset of vertices, so that $|\mathcal{V}_a^1| = \sum_{v\in\mathcal{V}_a} a_v$. Let $\mathcal{G}_a^1 = (\mathcal{V}_a^1, \mathcal{E}_a^1)$ be the corresponding unit-weight interval graph, where two vertices are adjacent if and only if their associated intervals overlap. By construction, imposing $a^\intercal x \leqslant b$ implies that the overlap graph induced by $\mathcal{G}_a^1$ is K_{b+1}-free. Since the Turán number for K_{b+1}-free interval graphs on $|\mathcal{V}_a^1|$ vertices is $\binom{|\mathcal{V}_a^1|}{2} - \binom{|\mathcal{V}_a^1|-b+1}{2}$, this writes as:

$$\sum_{(u,v)\in\mathcal{E}_a^1} z_u^v \leqslant \binom{|\mathcal{V}_a^1|}{2} - \binom{|\mathcal{V}_a^1| - b + 1}{2}. \tag{30}$$

For any two vertices $u, v \in \mathcal{V}_a$ of weights $a_u, a_v \neq 0$, the number of edges between their duplicates in $\mathcal{E}_a^1$ is $a_u \times a_v$, and for any vertex $u \in \mathcal{I}$, the number of edges between its duplicates in $\mathcal{E}_a^1$ is $\binom{a_u}{2}$. It follows that inequality (30) can be written as:

$$\sum_{(u,v)\in\mathcal{E}_a} a_u a_v . z_u^v + \sum_{u\in\mathcal{I}} \binom{a_u}{2} z_u^u \leqslant \binom{|\mathcal{V}_a^1|}{2} - \binom{|\mathcal{V}_a^1| - b + 1}{2}. \tag{31}$$

Given that any vertex $u \in \mathcal{I}$ always overlaps its duplicates, it follows that $z_u^u = 1$, yielding:

$$\sum_{(u,v)\in\mathcal{E}_a} a_u a_v . z_u^v \leqslant \binom{|\mathcal{V}_a^1|}{2} - \binom{|\mathcal{V}_a^1| - b + 1}{2} - \sum_{u\in\mathcal{V}_a : a_u \geqslant 2} \binom{a_u}{2}. \tag{32}$$

This formula applies to any inequality $a^\intercal \mathbf{x} \leqslant b$. In the special case of minimal and extended covers, the coefficients satisfy $a_v \in \{0, 1\}$ for all vertices v, and the construction reduces to the clique-forbidding inequalities we derived previously.

Example 3 (*cont.*). Consider $\mathbf{KP}^{\varphi,\bar{\varphi}}$ from Example 3 and the valid inequality $x_2 + x_3 + x_4 + 2x_5 \leqslant 3$. Applying the formula results in the clique-forbidding inequality

$$z_2^3 + z_2^4 + 2z_2^5 + z_3^4 + 2z_3^5 + 2z_4^5 \leqslant 6,$$

and yields the same result obtained in Example 3 by considering minimal and extended covers.

6.3. Integration in the branch-and-cut algorithm

The clique-forbidding inequalities derived from valid knapsack inequalities beyond minimal covers are generally not model-defining: the absence of violations in this family does not certify that the peak-power constraint is satisfied. In Example 3, activating operations $3, 4, 5$ simultaneously violates the peak limit and the minimal cover inequality, yet the two inequalities derived from the extended and lifted covers are satisfied. Accordingly, we retain the separation scheme of Section 5 and apply it on infeasible integer solutions using the values of the overlap variables z. Since minimal-cover inequalities are model-defining, they are always checked for violation. The remaining families of clique-forbidding inequalities are then tested after an infeasibility is detected. The resulting separation routine is summarized in Appendix 3.

7. Lazy cut selection

Consider Example 3 and an infeasible integer candidate solution where all five operations execute simultaneously. In this case, the three derived inequalities are violated. At each midpoint, multiple cuts may be violated, and this can occur across several midpoints in a schedule. This raises the question: *given several inequalities violated at an infeasible integer solution, which ones should be added?* Adding one inequality suffices to separate the current infeasible point, but doing so at every node could lead to a large enumeration tree and slow convergence. Conversely, adding all the inequalities could overburden subsequent LP relaxations. Furthermore, inequalities from minimal covers are more sparse, and thus more computationally advantageous for LP solvers (Gill et al., 1984). On the other hand, extended or lifted cover inequalities provide deeper cuts, but are relatively dense. This reflects the classic strength–density trade-off in cutting-plane algorithms (Balas et al., 1996) and motivates a cut-selection policy.

7.1. Cut selection measures

Cut selection methods aim to quantify the impact of a cut on the dual bound and, indirectly, on the evolution of the primal bound, using a variety of metrics (see Turner et al., 2023 and references therein). We denote by $a^\intercal \mathbf{x} \leqslant b$ a candidate cut and by $\bar{\mathbf{x}}$ the current infeasible integer solution. We consider the following cut evaluation measures, normalized to $[0, 1]$ so that larger values indicate better cuts:

- Efficacy: or cutoff-distance, measures the Euclidean distance between $\bar{\mathbf{x}}$ and its orthogonal projection onto the cut hyperplane. It is given by $\texttt{eff} := \frac{a^\intercal \bar{x} - b}{\|a\|}$.
- Sparsity: measures the number of non-zero coefficients in the cut, with fewer non-zeros being preferable, and is defined as $\texttt{spars} := 1 - \frac{|\{i \in [n] \| a_i \neq 0\}|}{n}$.
- Directed cutoff-distance: evaluates the distance between $\bar{\mathbf{x}}$ and the cut hyperplane in the direction of a primal solution. Let $\hat{\mathbf{x}}$ denote the incumbent solution vector. The directed cutoff-distance is given by $\texttt{dcd} := \frac{a^\intercal \bar{\mathbf{x}} - b}{|a^\intercal \mathbf{d}|}$, where $\mathbf{d} = \frac{\hat{\mathbf{x}} - \bar{\mathbf{x}}}{\|\hat{\mathbf{x}} - \bar{\mathbf{x}}\|}$.
- Objective parallelism: measures the degree of parallelism of the cut to the objective function. Let c denote the coefficient vector of the objective function, the parallelism is $\texttt{parall} := \frac{a^\intercal c}{\|a\|\|c\|}$.
- Orthogonality: measures the orthogonality of cuts with respect to one another. For two cuts $a_1^\intercal \mathbf{x} \leqslant b_1$ and $a_2^\intercal \mathbf{x} \leqslant b_2$, orthogonality is defined as $\texttt{ortho} := 1 - \frac{|a_1^\intercal a_2|}{\|a_1\|\|a_2\|}$.

7.2. Problem-specific metrics

The lazy cuts under consideration are all clique-forbidding inequalities. We derive metrics that are specific to this setting. Sparsity and efficacy are as previously defined and directly compare minimal, extended and lifted cover inequalities. The other metrics are adapted to exploit scheduling structure: they score a violated cut according to where the infeasibility occurs in the schedule, in particular with respect to the ToU periods and the power demands of the machines involved.

- Directed cutoff-distance: the computation of the direction vector is adjusted to consider the ToU period and the operations at which infeasibility is detected. This refinement uses the main processing duration variables $d_{j,m}^p$, since the overlap variables z are only indicators and do not localize where (in time) the violation occurs. Let $\mathcal{O}_{\texttt{inf}}$ denote the set of operations active at the infeasible midpoint, and let p be the corresponding ToU period. Let $\mathbf{d}_{\texttt{Pr}}$ be the projection of the direction vector onto these components. We define the adapted directed-cutoff distance as $\texttt{dcd} := \frac{a^\intercal \bar{\mathbf{x}} - b}{|\mathbf{d}_{\texttt{Pr}}|}$, where $\mathbf{d}_{\texttt{Pr}} = \frac{d^p_{\mathcal{O}_{\texttt{inf}}} - \bar{d}^p_{\mathcal{O}_{\texttt{inf}}}}{\|\hat{d} - \bar{d}\|}$.
- Objective parallelism: because overlap variables are not involved in the objective function, this measure is adapted to account for the ToU period and the power demands of the operations involved in the infeasibility. Let $\mathcal{O}_{\texttt{inf}}$ denote the set of operations active at the infeasible midpoint, and let p be the corresponding ToU period. The adapted measure is defined as $\texttt{parall} = \mu \cdot \frac{\min_k \kappa^k}{\kappa^p} +$

$(1-\mu)\cdot\frac{\min_{m\in\mathcal{M}}\varphi_m}{\text{avg}_{m\in\mathcal{O}_{\text{inf}}}\varphi_m}$, where μ is a mixing parameter and κ^p denotes the ToU cost scalar for p.

- Orthogonality: since overlap variables are defined for operation pairs, orthogonality is adapted to evaluate the intersection of operation sets. Let $\mathcal{O}_1$ and $\mathcal{O}_2$ denote the set of operations that describe two cuts. The adopted definition of orthogonality is $\texttt{ortho} = 1 - \frac{\text{Card}(\mathcal{O}_1\cap\mathcal{O}_2)}{\max(|\mathcal{O}_1|,|\mathcal{O}_2|)}$.

7.3. Cut filtering and scoring function

The overall quality of a cut is quantified using a weighted sum of the individual metric scores. The weights and the value of μ will be specified in the computational experiments. Since pairwise cut orthogonality is difficult to compute, we adopt a greedy parallelism-based approach for cut filtering (Achterberg, 2007). At each node, the score of each cut is computed with a default value of 1 for orthogonality. The highest-scoring cut is selected, and the orthogonality of each remaining cut is computed relative to the selected one. To ensure that the added cuts form an almost pairwise orthogonal set of hyperplanes, cuts failing a minimal orthogonality threshold are discarded. This process repeats until the available cuts are exhausted, or until a preset criterion—such as the number of selected cuts—is satisfied. The algorithm pseudocode is provided in Appendix 4.

8. Computational experiments

This section provides a detailed analysis of the conducted computational experiments to assess the performance of the proposed branch and cut (B&C) and its enhancements. It begins with an introduction to the benchmark instances used in the experiments, followed by a description of the computational study, the results obtained, and their analysis.

8.1. Instance generation

Various benchmark instances were used in this work. In addition to existing datasets, we propose newly generated benchmark sets by extending the established literature. We first provide a brief description of the benchmark instances introduced in Masmoudi et al. (2019) for the Jm|P$_{\max}$|TEC. These benchmarks are derived from two job-shop scheduling problem instances: *ft06* (Muth & Thompson, 1963), which involves 6 jobs and 6 machines (6×6), and *la04* (Lawrence, 1984) with 5 jobs and 10 machines (5×10). The authors adapted these instances to the problem under study by introducing additional parameters:

Machine power values: Five sets $(\varphi_m)_{m\in\mathcal{M}}$, drawn independently from the uniform distribution $\mathcal{U}[5,10]$.

Planning horizon values: Three different values $\lambda\cdot C^\star$, where $C^\star$ denotes the minimal makespan under a maximum power constraint, and $\lambda\in\{1.0,1.1,1.2\}$.

Peak power thresholds: Three thresholds, set as $\alpha\cdot\sum_{m\in\mathcal{M}}\varphi_m$ where $\alpha\in\{0.7,0.9,1.0\}$. We focus on instances where the peak-power constraint is binding, i.e., $\alpha\in\{0.7,0.9\}$.

Time-of-Use profiles: The ToU tariffs follow French electricity price profiles, alternating off-peak and on-peak periods with an off:on length ratio of 1:2.

Overall, this parameterization yields 90 instances, of which we consider 60 in our experiments. Building on the existing generation scheme, we introduce additional benchmark families by combining the following features:

Job-shop scheduling instances: We generate new JSSP instances with a first set of 7×7 and 6×8 configurations, and a second set including instances of dimensions 6×10, 7×9, and 8×8. Consistent with classical job-shop benchmarks, job sequences are derived from random permutations, and processing times are integers drawn uniformly from [4, 99] (e.g. Adams et al., 1988).

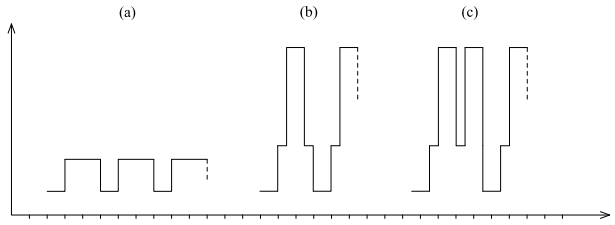


Fig. 3. (a) off-on, (b) off-mid-on and (c) off-2x[mid-on] peak ToU profiles.

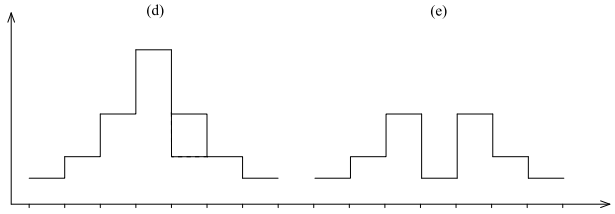


Fig. 4. (d) pyramidal and (e) valley-pyramidal ToU profiles.

Machine power values and peak power limit: We consider four distinct power value sets for the new families of instances, inspired by knapsack sets known in the literature: the three-coefficient knapsack (KP3), the weakly-super-increasing knapsack (WSI), the small and large coefficient knapsack (sL), and the arithmetic sequence knapsack with common difference of 1 (1-AS). A complete overview of knapsack polytopes can be found in Hojny et al. (2020).

Consider machine power values $(\varphi_m)_{m\in\mathcal{M}}$. For a given $\alpha < 1$, the peak power threshold is chosen as the closest integer value to $\alpha\sum_{m\in\mathcal{M}}\varphi_m$. These values are drawn from the discrete set of attainable values $\mathcal{W}$, defined as

$$\mathcal{W} := \Bigg\{\overline{\varphi} = \sum_{i\in\mathcal{M}}\varphi_i x_i \,:\, \max_{i\in\mathcal{M}}\varphi_i \leqslant \overline{\varphi} \leqslant \sum_{i\in\mathcal{M}}\varphi_i - \min_{i\in\mathcal{M}}\varphi_i, x_i\in\{0,1\}, \forall i\in\mathcal{M}\Bigg\}.$$

We compute $\mathcal{W}$ using a dynamic programming procedure analogous to the one used to enumerate normal patterns in cutting stock problems (Christofides & Whitlock, 1977). For each generated instance, we also compute the makespan $C^\star$ under the peak power threshold $\overline{\varphi}$. Appendices 5.1 and 5.2 detail the computation of $\mathcal{W}$ and $C^\star$.

New Time-of-Use profiles: In addition to the alternating off- and on-peak ToU profile of Masmoudi et al. (2019), we consider two additional profiles from real-world scenarios: a three-level ToU profile featuring an additional mid-peak period (Zhang et al., 2014), denoted as off-mid-on, and a profile with two additional alternating on- and mid-peak periods denoted as off-2x[mid-on] (Ding et al., 2016). Fig. 3 illustrates a representative segment of each of the three profiles.

To complete our set, we introduce (see Fig. 4) two additional profiles: a pyramidal ToU, in which periods form a pyramid-like structure; and a valley-pyramidal ToU, featuring a low-price valley in the middle. These profiles aim to explore combinatorial behaviors and, to the best of our knowledge, do not reflect realistic tariff schedules.

The complete set of features considered for the benchmark instances used in this work—including dimensions, machine power sets, and ToU profiles—is summarized in Table 3. Further details on the knapsack sets, scheduling-horizon values, and ToU profiles are provided in Appendices 5.3, 5.4, and 5.5, respectively.

The benchmark instances are designed to explore several complementary sources of difficulty: instance size (number of operations and processing-time ranges), planning horizon length (expressed relative to the optimal makespan under the peak-power constraint), and machine power distributions together with the peak-power thresholds, which determine the underlying knapsack structure.

Table 3
A summary of the considered features for the benchmark instances.

Feature	Existing instances	New instances
$\lvert\mathcal{J}\rvert \times \lvert\mathcal{M}\rvert$	6×6, 5×10	6×8, 7×7, 6×10, 7×9, 8×8
Power values $(\varphi_m)_{m\in\mathcal{M}}$	5 sets from $\mathcal{U}$[5,10]	KP3, sL, WSI, 1-AS
ToU profile	Off-on peak	Off-mid-on, off-2x[mid-on], pyramidal, valley-pyramidal

8.2. Results

This section presents an analysis of computational results to evaluate and compare the performance of B&C and its successive enhancements. We begin with an initial comparison against the model of Masmoudi et al. (2019), which serves as a reference baseline. We then assess the impact of the derived valid inequalities on tightening linear relaxations and improving tree search efficiency, identifying their most effective combination. Building on this, we examine the performance of clique-interdiction constraints derived from the facet-inducing (non-trivial) inequalities describing the peak power polytope. We additionally evaluate the effectiveness of the adapted cut selection framework under different configurations of cut scoring parameters. Finally, we investigate the impact of energy-related features on the performance of the best-performing algorithm.

All MILPs were implemented using `JuMP` (Lubin et al., 2023) and solved with the commercial solver `Gurobi 11.0` (Gurobi Optimization, LLC, 2024). The branch-and-cut algorithms were developed in `Julia v1.10` using solver-dependent callbacks. Experiments were conducted with a CPU time limit of one hour per instance, running on a single core of an Intel Xeon Gold 6132 at 2.60 GHz. All `Gurobi` parameters are set to their default settings.

Due to the small dimension of the peak-power polytope (equal to the number of machines), its complete polyhedral description is precomputed once, in negligible time (under 5 ms), using `Polymake` (Gawrilow & Joswig, 2000). The resulting facet-inducing inequalities are then used to formulate inequalities (25) and to derive the clique-forbidding constraints in inequalities (32). In the following tables, T/(%) reports the CPU time when the instance is solved to optimality; otherwise, it reports the final optimality gap at the time limit (if above the default optimality tolerance). A missing entry indicates that no feasible solution was found within the time limit. We denote by `dim` the size ($|\mathcal{J}|\text{x}|\mathcal{M}|$) of the JSSP problem. #cols and #rows refer to the number of columns and rows of a model, both measured prior to presolve by `Gurobi`. $\%^{\texttt{root}}_{\texttt{bks}}$ denotes the relative gap between the value of the root LP relaxation and the best-known solution for the instance, and (#n) denotes the number of B&B nodes explored. $\#\mathbf{x}^\star$ and $\#\overline{\mathbf{x}}$ respectively denote the number of instances solved to optimality and those for which at least one feasible solution was found under the time limit.

To reduce sensitivity to outliers, we compute averages using the shifted geometric mean, defined as $\left(\prod_{i=1}^{n}(t_i+s)\right)^{1/n}-s$, where $t_1,\ldots,t_n$ are the values and s is the shift. The shifted geometric mean is commonly adopted in mixed-integer programming benchmarking to provide robust aggregate comparisons and to limit the influence of outliers (Berthold & Hendel, 2021). Unless stated otherwise, the shift values are $s=100$ for the number of nodes, $s=10$ for CPU time and cut-related averages, $s=1$ for #cols and #rows, and $s=0.1$ for gaps and related measures.

8.2.1. Comparison with the reference model and evaluation of a first subset of valid inequalities

In Table 4, we compare the performance of the reference time-indexed model of Masmoudi et al. (2019), referred to as IP2, against two proposed variants $\texttt{B\&C}_1$ and $\texttt{B\&C}_2$, of the branch-and-cut algorithm based on minimal covers. The benchmark instances used for the comparison were also introduced in Masmoudi et al. (2019). $\texttt{B\&C}_1$ uses formulation (MIP) and inequalities (13)–(18), where inequalities (13) and (14) were shown in Felloussi et al. (2025) to significantly improve the root relaxation of a period-indexed formulation for the Jm||TEC. $\texttt{B\&C}_2$ additionally includes inequalities (25) and (26), which account for the capacity aspect. The two rightmost columns show the ratio of variables (#cols) and constraints (#rows) between IP2 and $\texttt{B\&C}_2$. Results are aggregated over the power-value sets, which are drawn from the same uniform distribution, and each row reports averages over five instances.

As observed in prior work, IP2 has a strong root relaxation but scales poorly as the time horizon increases. This is evident in the 5×10 instances, where the time to prove optimality increases and fewer instances are solved within the time limit. $\texttt{B\&C}_1$ performs well for small horizons ($\lambda=1.0$) but struggles otherwise, including the 6×6 instances. Its weaker linear relaxation leads to large enumeration trees and failure to prove optimality within time limits, highlighting the need for a stronger linear relaxation.

$\texttt{B\&C}_2$ has a tighter root relaxation gap and becomes competitive with IP2 while maintaining a more compact model, as shown by its better model size ratios. The gains are most visible on larger horizons. Further, $\texttt{B\&C}_2$ performs well across all instances, solving the 6×6 instances in under 5 s on average and the 5×10 instances in under 90 s. In contrast, IP2 averages roughly 2000 s and often fails to prove optimality on the 5×10 instances.

In the tables of the following sections, average CPU times account for the one-hour limit when optimality is not reached. Final optimality gaps are not reported and compared, as they are consistently similar across variants. The primary evaluation criteria are the number of instances solved to optimality and the corresponding average CPU time.

8.2.2. Comparison of other valid inequalities

In Table 5, we conduct an ablation study to quantify the contribution of the additional cut families on top of $\texttt{B\&C}_2$. We denote the implied bound (20)–(23) and duration integrality (24) inequalities by `ib`, the generalized non-preemption inequalities (19) by `gnp`, and the flow cover inequalities (28) by `fc`. We compare $\texttt{B\&C}_2$ with variants $\texttt{B\&C}_{\texttt{all-cut}}$ that include all cut families except one among $\{\texttt{ib},\texttt{gnp},\texttt{fc}\}$, and with $\texttt{B\&C}_{\texttt{all}}$, which includes all inequalities. All inequalities are added statically, i.e., they are part of the initial model.

The benchmark set for these experiments includes the 30 existing 5×10 instances, and 24 new instances of each size 6×8 and 7×7, with machine power values of types KP3, sL, WSI, and 1-AS. The peak power values are selected from $\mathcal{W}$ with $\alpha\in\{0.5,0.6\}$, and λ varies in $\{1.05,1.1,1.2\}$. The case $\lambda=1.0$ is excluded, as it results in no feasible solutions for all variants. The tests assume the off-on peak ToU profile previously used, and results are averaged over φ and α values.

The rightmost column reports only the constraint-number ratio between $\texttt{B\&C}_{\texttt{all}}$ and $\texttt{B\&C}_2$, as they have the same number of variables. None of the additional cut families improves the root relaxation of $\texttt{B\&C}_2$; we therefore omit this metric. The implied bound cuts appear to have the largest impact: excluding them leads to fewer instances solved to optimality and the largest increase in CPU time. In contrast, flow cover inequalities have little effect, while excluding generalized non-preemption cuts reduces computing time but results in three fewer optimal solutions. Overall, adding all proposed cuts yields a modest improvement over $\texttt{B\&C}_2$, reducing the average time to optimality and solving one additional instance within the time limit. We attribute these gains to improved pruning of the B&B tree through strong subtree dual bounds, which outweighs the slower node relaxations induced by the additional cut families, despite the resulting model having roughly twice as many constraints as $\texttt{B\&C}_2$. In the following experiments, B&C denotes $\texttt{B\&C}_{\texttt{all}}$.

Beyond the impact of valid inequalities, we note that decreasing α increases the problem dimension by extending the scheduling horizon. This effect is reflected in the computational results of the 5×10 instances: compared to the cases $\alpha\in\{0.7,0.9\}$ reported in Table 4, two fewer instances are solved to optimality within the time limit for $\alpha\in\{0.5,0.6\}$, and the average time required to reach optimality increases.

Table 4
Comparison of two variants B&C$_1$ and B&C$_2$, and the time-indexed formulation IP2 of Masmoudi et al. (2019).

	IP2			B&C$_1$			B&C$_2$			Relative ($\frac{\text{B\&C}_2}{\text{IP2}}$)	
dim/α/λ	T/(%)	$\%^{\text{root}}_{\text{bks}}$	#x*/$\bar{x}$	T/(%)	$\%^{\text{root}}_{\text{bks}}$	#x*/$\bar{x}$	T/(%)	$\%^{\text{root}}_{\text{bks}}$	#x*/$\bar{x}$	#rows	#cols
6×6/0.7/1.0	3.3	0.27%	5/5	1.4	2.13%	5/5	**1.1**	0.27%	**5/5**	0.93	0.79
6×6/0.7/1.1	5.1	0.35%	5/5	(0.42%)	2.24%	0/5	**4.7**	0.35%	**5/5**	0.93	0.78
6×6/0.7/1.2	12.4	0.06%	5/5	(1.04%)	2.70%	0/5	**4.0**	0.06%	**5/5**	0.85	0.71
6×6/0.9/1.0	0.3	0.21%	5/5	0.1	1.00%	5/5	**0.1**	0.18%	**5/5**	0.95	0.81
6×6/0.9/1.1	0.8	0.30%	5/5	18.1	1.10%	5/5	**0.6**	0.29%	**5/5**	0.95	0.79
6×6/0.9/1.2	**1.0**	0.33%	5/5	28.3	1.13%	5/5	**1.0**	0.33%	**5/5**	0.87	0.72
5×10/0.7/1.0	–	0.00%	0/0	(3.53%)	3.42%	0/5	**89.8**	0.00%	**5/5**	0.10	0.07
5×10/0.7/1.1	2122.6/(0.06%)	0.00%	3/5	(3.98%)	3.93%	0/5	**16.9**	0.00%	**5/5**	0.10	0.07
5×10/0.7/1.2	2046.2	0.00%	5/5	(3.96%)	3.93%	0/5	**13.2**	0.00%	**5/5**	0.09	0.07
5×10/0.9/1.0	2756.5	0.00%	1/1	(1.22%)	1.21%	0/5	**49.2**	0.00%	**5/5**	0.12	0.09
5×10/0.9/1.1	2402.5	0.01%	2/2	(1.15%)	1.18%	0/5	**9.6**	0.01%	**5/5**	0.11	0.08
5×10/0.9/1.2	1671.4/(0.32%)	0.00%	4/5	(1.15%)	1.23%	0/5	**4.0**	0.00%	**5/5**	0.10	0.08

Table 5
Comparison of valid inequalities over the new instances.

	B&C$_2$			B&C$_{\text{all-ib}}$			B&C$_{\text{all-gnp}}$			B&C$_{\text{all-fc}}$			B&C$_{\text{all}}$			($\frac{\text{B\&C}_{\text{all}}}{\text{B\&C}_2}$)
dim/λ	T	#n	#x*	T	#n	#x*	T	#n	#x*	T	#n	#x*	T	#n	#x*	#rows
5×10/1.05	82.9	8510	9	**71.3**	5745	**10**	95.5	8111	9	88.3	6706	9	125.5	8649	9	
5×10/1.1	67.0	5888	9	88.1	6074	10	51.9	3434	9	54.4	2717	9	**50.2**	2331	**10**	2.17
5×10/1.2	120.4	9385	9	65.6	2860	9	**62.9**	2559	**9**	95.3	3435	9	86.2	3145	9	
6×8/1.05	66.7	8627	8	55.9	5033	7	70.1	6032	7	56.4	4213	8	**33.6**	2301	**8**	
6×8/1.1	51.0	5152	8	69.7	5824	7	48.2	3010	7	**41.3**	3658	**8**	55.1	2760	8	1.93
6×8/1.2	34.8	3584	8	41.2	3108	8	23.1	1491	8	19.6	1190	8	**18.1**	1080	**8**	
7×7/1.05	385.7	51K	6	456.6	51K	5	**236.1**	26K	**6**	283.3	30K	6	330.8	35K	6	
7×7/1.1	119.4	19K	7	159.8	22K	6	**83.0**	8569	**7**	114.8	14K	7	155.8	18K	7	1.83
7×7/1.2	60.1	8984	7	64.6	6221	7	65.9	5651	7	70.5	6343	7	**54.1**	4297	**7**	
All (78)	86.16	9613	71	89.5	7403	69	**69.55**	5163	69	73.55	5309	71	74.81	4783	**72**	2.02

8.2.3. Comparison of the peak-power constraint separation methods

The results in Table 6 evaluate the performance of three variants that differ in how lazy cuts are generated. B&C$_{\text{mcv}}$ adds only clique-forbidding inequalities derived from minimal covers. B&C$_{\text{mcv+fi}}$ checks minimal covers first and, when violations are detected, also tests facet-inducing inequalities, i.e., extended (Section 6.1) and general lifted covers (Section 6.2), of the peak power polytope. Variant B&C$_{\text{mcv+fi_prio}}$ prioritizes facet-inducing inequalities: if any are violated, only the corresponding cuts are added; otherwise, the procedure falls back to minimal covers to verify infeasibility. All variants share the same correctness guarantee described in Section 5. Whenever a peak-power violation is detected at a candidate integer solution, the separation procedure adds at least one violated clique-forbidding inequality. The variants differ only in how many and which violated inequalities are added at each separation call. Minimal covers are always considered, since inequalities from extended and general lifted covers are not model-defining.

The benchmark set for these experiments is derived from 6×10, 7×9, and 8×8 JSSP instances; machine power sets of type KP3, sL, WSI, and 1−AS; a peak power threshold corresponding to $\alpha = 0.6$; and scheduling horizon where $\lambda \in \{1.05, 1.1, 1.2\}$. We also consider two ToU pricing profiles of the same off-/on-peak structure, differing only in the cost ratio between the off- and on-peak periods. This variation is introduced solely to enlarge the benchmark set and improve statistical coverage at this stage; we do not analyze the impact of ToU pricing here. The metric #cuts will refer to the number of added lazy cuts, with %add indicating the proportion among minimal covers (mcv), extended covers (ext), and general lifted covers (lci), reported as %(mcv)/%(ext)/%(lci). Results in the first part of the table are aggregated over λ and the two ToU profiles; the remaining rows report the corresponding subsets.

For each instance, the cut-type proportions added by B&C$_{\text{mcv+fi_prio}}$ provide an empirical proxy for the cover families that describe the underlying knapsack polytope induced by the machine powers. Since B&C$_{\text{fi_prio}}$ stops at the first violated inequality, a larger share of a family indicates that its inequalities most often yield the first separation. The comparison is relative, as minimal cover (clique-forbidding) inequalities are always checked and thus form a baseline. For example, for the sL/6×10 and 1−AS/8×8 instances, the knapsack polytopes are predominantly described by lifted-cover inequalities, which is reflected in their higher observed proportions, whereas the WSI instances are essentially described by minimal-cover inequalities.

While B&C$_{\text{mcv+fi}}$ adds more lazy cuts per separation round, the resulting strengthening offsets the additional model load, solving more instances to optimality in lower CPU time on average. This effect is more pronounced on sets where the knapsack polytope is described by extended and general lifted cover inequalities, such as 1−AS and sL for the 6×10 JSSP instance. For WSI sets, which are fully characterized by minimal-cover inequalities, performance differences across the three variants are negligible. The remaining variations are not algorithmic: on these sets, inequality families coincide, and the variants mainly differ in the order in which violated inequalities are detected and added. In particular, B&C$_{\text{mcv+fi}}$ and B&C$_{\text{mcv+fi_prio}}$ first scan for violated cuts derived from facet-inducing inequalities, which for WSI are all minimal covers, but they return violated cuts in different orders, leading to different cut insertion orders in our implementation. Since Gurobi stores lazy cuts in a pool and may use them only later, this order may have influenced the results. On the KP3 set, where facet-inducing inequalities are predominantly minimum and extended covers, adding only minimal-cover inequalities appears to be the better strategy. For instance, while the average performance of B&C$_{\text{mcv+fi_prio}}$ is overall comparable to that of B&C$_{\text{mcv}}$, it notably underperforms on the KP3 set.

Overall, the knapsack structure induced by the machine-power values helps predict which variant performs best, though instance size and cut sparsity play a role. For example, while sL/6×10 and 1−AS/8×8 exhibit similar cut-type proportions, B&C$_{\text{mcv+fi_prio}}$ performs best on the former but worst on the latter. We also observe that in 1−AS/8×8,

Table 6
Comparison of clique-forbidding inequalities over the new instances.

	B&C$_{mcv}$			B&C$_{mcv+fi}$				B&C$_{mcv+fi_prio}$			
Subset	T	#x*	#cuts	T	#x*	#cuts	%add	T	#x*	#cuts	%add
sL/6×10	64.4	6	643	64.8	6	772	88/1/11	**45.7**	**6**	573	86/1/13
sL/7×9	**75.5**	**6**	526	112.0	6	709	95/2/3	82.3	6	589	94/2/4
sL/8×8	2536.7	2	568	**1795.3**	**2**	506	98/2/0	2574.5	2	643	97/3/0
sL (all)	242.9	14	577	244.43	14	652	94/1/5	**226.9**	**14**	601	93/1/6
WSI/6×10	**97.9**	**6**	138	109.5	6	162	100/0/0	109.5	6	162	100/0/0
WSI/7×9	1024.3	4	318	**895.6**	**4**	357	100/0/0	**895.6**	**4**	357	100/0/0
WSI/8×8	678.5	6	156	**597.4**	**6**	153	100/0/0	**597.4**	**6**	153	100/0/0
WSI (all)	415.1	16	190	**393.5**	**16**	207	100/0/0	**393.5**	**16**	207	100/0/0
KP3/6×10	**108.3**	**6**	583	113.7	6	724	95/4/1	114.7	5	544	94/5/1
KP3/7×9	TL	0	2603	TL	0	2799	95/4/1	TL	0	2333	95/4/1
KP3/8×8	**52.0**	**6**	356	66.8	6	418	99/1/0	123.8	6	435	98/1/1
KP3 (all)	**288.1**	**12**	817	314.9	12	949	96/3/1	382.0	12	822	96/3/1
1-AS/6×10	85.0	6	659	**55.6**	**6**	573	93/1/6	83.2	6	701	92/1/7
1-AS/7×9	454.4	6	901	**235.2**	**6**	887	91/2/7	317.7	6	866	91/2/7
1-AS/8×8	1304.3	3	1144	**1189.3**	**4**	1155	86/1/13	1348.9	3	1017	89/1/10
1-AS (all)	377.1	15	879	**258.25**	**16**	838	90/1/9	336.2	15	851	91/1/8
All (6×10)	87.5	24	506	**82.3**	**24**	558	94/2/4	83.8	23	495	95/2/3
All (7×9)	610.4	16	1087	**549.2**	**16**	1188	95/2/3	550.8	16	1036	92/4/4
All (8×8)	604.8	17	556	**553.8**	**18**	558	95/1/4	720.9	17	562	94/2/4
All [#inst=72]	323.7	57	529	**297.6**	**58**	571	95/2/3	327.5	56	543	94/2/4

B&C$_{mcv+fi_prio}$ adds a higher share of mcv cuts. This is likely because a longer time to prove optimality is required, and as the gap narrows, violations of ext and lci inequalities may decrease, leaving mcv as the dominant cut type.

Overall, while B&C$_{mcv+fi}$ tends to perform best on average, these results suggest that adaptive cut selection could further improve performance by tailoring the separation strategy to instance characteristics. Here, cut selection is driven by the combinatorial origin of the inequalities; in the next section, we investigate cut selection based on general cut-quality metrics. In the following experiments, the baseline B&C refers to B&C$_{mcv+fi}$.

8.2.4. Evaluation of lazy cut selection

Table 7 evaluates a lazy cut selection strategy applied to cuts derived from minimal cover and facet-inducing inequalities of the peak power polytope, and based on the problem-specific measures considered for this study. The selection process considers all the measures discussed in Section 7: efficacy/cutoff-distance (eff), directed-cutoff-distance (dcd), sparsity (spars), objective parallelism (parall), and cut orthogonality (ortho). We test the base configuration B&C$_{base}$—where all scoring weights are assigned value 1—against the variants B&C$_{no_spars}$, B&C$_{no_dist}$ and B&C$_{no_parall}$ where the weights associated with sparsity, distance (both eff and dcd) and objective parallelism are set to 0 respectively. We do not test for orthogonality since its importance is well-studied in the literature. Cuts are filtered using the parallelism-based approach discussed in Section 7.3, with a minimum orthogonality threshold of 0.8. The number of added cuts is limited to 10 when an incumbent is available and to 30 otherwise. Cut selection is only performed when the node gap exceeds 0.1%, and otherwise all violated cuts are added. The mixing parameter for parallelism μ is set to 0.25. In the remainder of this section, %filt will refer to the proportion of discarded cuts, i.e., violated cuts that are removed by the filtering step after selection. We use identical test instances to the ones in the experiments of Section 8.2.3; results are reported by KP type and aggregated within each type over the ToU profiles and λ.

The results in Table 7 show that disabling either sparsity (B&C$_{no_spars}$) or objective parallelism (B&C$_{no_parall}$) most often yields the largest increase in time to prove optimality, whereas disabling distance-based measures (B&C$_{no_dist}$) has little effect across the testbed. All variants solve the same number of instances to optimality (#x* = 58), and the optimality gaps at the time limit are very similar; we therefore focus on time to optimality comparisons.

We observe that sparsity and objective parallelism are key selection metrics in this setting. When either measure is removed, the selected cuts shift toward ext and lci inequalities whenever available, as reflected by the cut-type proportions. This shift is most visible on the 1-AS and sL sets, whose knapsack polytopes are largely described by ext and lci inequalities. In contrast, on WSI instances there is no significant performance difference across variants. This matches the earlier cut-type analysis: for WSI, the violated inequalities all stem from minimal covers and therefore have similar properties, leaving little leverage for an additional selection layer.

Distance-based measures have limited impact in this lazy-cut setting. When violation (via eff/dcd) is not used, the procedure tends to prioritize mcv cuts, leading to behavior close to B&C$_{mcv}$, in line with the previous cut-type results. This outcome runs against the initial motivation for directed-cutoff distance, namely that cuts aligned with an incumbent would be especially effective when separating incumbent solutions.

On the 1-AS/8×8 instances, we also see that the filtered cuts are dominated by mcv inequalities relative to other families, consistent with the earlier observation that, deeper in the search, violated ext and lci inequalities become less prevalent and minimal covers dominate. This is mirrored by B&C$_{no_dist}$, which requires substantially more time to reach optimality on these instances.

Overall, the results suggest that relying primarily on violation-based measures is not sufficient in this context, and that sparsity- and parallelism-based criteria play a central role in effective lazy-cut selection. The basic configuration B&C$_{base}$ slightly outperforms adding all cuts (B&C$_{mcv+fi}$); further gains may follow from tuning the selection weights. In what follows, B&C$_{base}$ is retained as the main algorithm.

8.2.5. Sensitivity to ToU profiles

In Table 8, we examine the impact of energy-related features using our best-performing algorithm. We consider two additional variants of the reference off-on peak ToU profile, which differ in off-/on-peak length ratio (1:1 and 2:1, versus 1:2 in the profile used so far). We further evaluate the algorithm on four additional ToU profiles: off-mid-on peak, off-2x[mid-on] peak, pyramidal and valley-pyramidal. All profiles are defined so that the number of periods is comparable to that of the reference off-on (1:2) profile (see Appendix 5.5 for details).

Table 7
Comparison of different lazy cut selection configurations.

	$B\&C_{no_spars}$			$B\&C_{no_dist}$			$B\&C_{no_parall}$			$B\&C_{base}$		
KP/dim	T	%add	%filt	T	%add	%filt	T	%add	%filt	T	%add	%filt
sL/6×10	**36.5**	90/1/9	85/1/14	45.2	97/0/3	82/2/16	69.2	95/1/4	82/2/16	42.5	96/1/3	84/2/14
sL/7×9	127.5	95/2/3	91/3/6	111.8	97/1/2	89/4/7	95.6	96/1/3	91/3/6	**83.7**	95/1/4	92/3/5
sL/8×8	2621.4	97/3/0	98/2/0	**2246.4**	99/1/0	97/3/0	2365.6	97/3/0	99/1/0	2255.8	97/3/0	98/2/0
WSI/6×10	55.3	100/0/0	100/0/0	57.1	100/0/0	100/0/0	67.7	100/0/0	100/0/0	**46.3**	100/0/0	100/0/0
WSI/7×9	1336.1	100/0/0	100/0/0	**1204.5**	100/0/0	100/0/0	1320.9	100/0/0	100/0/0	1483.1	100/0/0	100/0/0
WSI/8×8	553.8	100/0/0	100/0/0	**535.1**	100/0/0	100/0/0	640.6	100/0/0	100/0/0	595.9	100/0/0	100/0/0
KP3/6×10	158.0	94/6/0	97/3/0	177.3	97/3/0	92/7/1	184.2	94/6/0	97/3/0	**154.4**	94/6/0	97/3/0
KP3/7×9	TL	–	–	TL	–	–	TL	–	–	TL	–	–
KP3/8×8	156.1	99/1/0	100/0/0	78.0	99/1/0	98/2/0	**60.7**	98/2/0	100/0/0	70.0	98/2/0	100/0/0
1-AS/6×10	157.5	89/1/10	79/2/19	89.4	99/0/1	87/2/11	107.7	97/0/3	88/3/9	**43.8**	95/1/4	86/3/11
1-AS/7×9	667.1	95/1/4	92/1/7	504.4	98/0/2	89/2/9	718.6	97/1/2	87/2/11	**353.0**	96/1/3	89/2/9
1-AS/8×8	2225.8	95/1/4	91/1/8	851.1	94/0/6	81/2/17	1064.0	89/1/10	75/2/23	**783.0**	92/0/8	77/2/21
All (#$\mathbf{x}^*$=58)	398.0	95/2/3	93/2/5	321.6	98/1/1	91/3/6	357.7	95/2/3	92/2/6	**284.0**	96/2/2	92/2/6

We use the same machine power value sets and JSSP instances as in Section 8.2.3, with a scheduling horizon coefficient of $\lambda = 1.2$ and a peak power threshold corresponding to $\alpha = 0.6$. Additional runs were performed with a time limit of 3TL (10 800 s) to evaluate the algorithm under longer runtimes. To complement $\%^{root}_{bks}$ for $B\&C_{base}$, we also report the relative difference between the root LP relaxation value of $B\&C_{base}$ and that of the time-indexed model IP2 (Masmoudi et al., 2019), measured with respect to IP2. We denote this quantity by Δ^{root}_{IP2}, where positive values indicate a stronger root relaxation for $B\&C_{base}$. The metric gap reports the final optimality gap at the corresponding time limit (TL or 3TL) for instances not solved to optimality. When aggregated, gap values are averaged only over instances not solved to optimality (optimal runs are excluded and not counted with a zero gap). Finally, $|\Delta^{bound}_{TL\to 3TL}|$ reports the relative change (in %) of the primal and dual bounds from TL to 3TL, computed with respect to their values at TL and reported in absolute values. These metrics help evaluate the impact of the ToU profiles of the branch-and-cut algorithm and provide insight into the quality and evolution of bounds across profiles. Gap-related measures are averaged with shift $s = 0.1$. Other symbols are defined as before, and each row reports results aggregated over the specified knapsack types and JSSP instances; detailed results are provided in Appendix 6.

The results highlight notable variations in performance across the different ToU profiles. While feasible solutions are found for all tested instances, the number of optimal solutions, the final gaps for instances not solved to optimality, and the number of explored nodes at TL vary markedly across profiles.

Across ToU profiles, the values of $\%^{root}_{bks}$ are often close to the gaps reported at TL, indicating that the dual bound improvement is not significant after cut separation rounds and global dual bound updates after branching, despite exploring 5×10^4 nodes on average. This is consistent with the relatively small bound variations between TL and 3TL, as summarized by $|\Delta^{bound}_{TL\to 3TL}|$. Moreover, Δ^{root}_{IP2} is positive across all tested profiles, showing that the root relaxation of $B\&C_{base}$ dominates that of the time-indexed formulation IP2. This supports the effectiveness of the proposed valid inequalities in strengthening dual bounds.

Extending the time limit from TL to 3TL yields four additional optimal solutions, but only marginal gap reductions on the remaining instances not solved to optimality. Across the ToU profiles, the bound-variation metrics show that the primal bound improves more than the dual bound between TL and 3TL, indicating that most post-TL progress comes from incumbent refinement. Combined with the root-relaxation comparison against IP2, this suggests that performance is primarily limited by the ability to generate improving solutions over time rather than by bound strength.

With the alternative off-on peak ratios (2:1 and 1:1), the algorithm achieves strong dual and primal bounds, with both root and final gaps below 0.35% on average. However, it proves optimality for only 3 of the 12 instances in each variant. A plausible explanation is that these profiles allocate a larger total duration to the lowest price (off-peak) regime relative to the scheduling workload than the previously considered (1:2) ratio case. Many schedules can then place most processing in cheap periods, leading to more solutions with similar objective values. This makes the remaining gap harder to close even when tight dual bounds are obtained early, and it is consistent with the smaller dual-bound progress observed from TL to 3TL.

A related difficulty arises when tariff patterns become more intricate. Among all profiles, the off-mid-on profile is the most challenging: it is the only one with no optimal solutions within TL, and it performs worst in terms of gap values, explored nodes, and root relaxations. Relative to the off-on variant, introducing an intermediate tariff level empirically makes it harder to obtain good incumbents (relative to TL), with most bound improvement being driven by the primal side throughout the run. The pyramidal-based profiles show a similar tendency, suggesting that profiles with more complex price patterns are more demanding for incumbent improvement.

In summary, while the algorithm performs consistently across the other instance features, the ToU profile remains the dominant driver of performance variation. Numerically, the profiles mainly differ in the relative lengths and prices of their tariff periods. At the same time, the intuition on near-equivalent solutions does not explain all comparisons; for instance, the valley-pyramidal profile contains one additional lowest-cost period compared to the pyramidal profile, yet its performance is slightly better. Overall, the interaction between tariff structure and algorithmic behavior appears multifaceted and is not captured by a single simple profile descriptor.

Lastly, to put our results in perspective, we also attempted to solve the same instances with the disaggregated version of IP2, which is known to yield even stronger LP relaxations (Artigues, 2017), but all runs terminated at the root node due to out-of-memory errors. The aggregated IP2 model also failed to produce feasible solutions within TL, likely because its large size slows down LP solves and limits the effectiveness of LP/MIP-based primal heuristics.

9. Conclusion

Energy-aware scheduling has become a central topic as industrial systems increasingly balance operational efficiency with sustainability objectives. In the combinatorial optimization literature, energy-related scheduling problems in manufacturing systems are most often addressed through matheuristic approaches tailored to specific industrial settings. In this work, we study the Job-Shop Scheduling Problem (JSSP) with energy considerations, minimizing total energy costs under a Time-of-Use (ToU) pricing scheme and a peak power limit. Our objective is primarily methodological: we investigate how energy constraints

Table 8
Comparison of the B&C$_{base}$ algorithm on different ToU profiles.

			Root		TL				3TL		$\lvert\Delta^{bound}_{TL\to 3TL}\rvert$	
ToU profile	#inst	#$\bar{x}$	$\%^{root}_{bks}$	Δ^{root}_{IP2}	T	#n	#$x^\star$	gap	#$x^\star$	gap	Dual	Primal
off-on (1:2)	24	24	0.15%	+0.01%	294.1	12K	18	0.31%	18	0.17%	0.03%	0.06%
off-on (2:1)	12	12	0.35%	+0.13%	2632.7	83K	3	0.34%	3	0.32%	0.01%	0.03%
off-on (1:1)	12	12	0.26%	+0.09%	2782.8	28K	3	0.35%	3	0.30%	0.01%	0.03%
off-mid-on	12	12	5.77%	+1.05%	TL	99K	0	4.89%	2	3.56%	0.19%	0.38%
off-2x[mid-on]	12	12	4.25%	+1.19%	3507.6	60K	1	3.05%	1	2.61%	0.15%	0.10%
pyr	12	12	6.57%	+2.87%	3021.4	79K	1	3.98%	2	3.53%	0.18%	0.30%
val-pyr	12	12	5.73%	+1.92%	2919.6	61K	1	3.25%	2	2.78%	0.16%	0.17%

and ToU pricing affect the structural and algorithmic properties of a classical scheduling problem, chosen for its generality and prominence in the scheduling and combinatorial optimization literature.

To address this problem, denoted Jm|P_{max}|TEC in the three-field notation, we propose a compact period-indexed MILP formulation that enforces the peak power limit through clique-forbidding constraints. These constraints are model-defining and are derived from the minimal covers of the associated knapsack set. To handle their exponential number, we embed the formulation in a branch-and-cut framework that dynamically separates violated clique-forbidding constraints at infeasible integer solutions. Starting from this baseline formulation, we strengthen the root relaxation by introducing several families of additional valid inequalities. These inequalities significantly tighten the formulation while preserving its compactness, allowing the resulting approach to outperform state-of-the-art time-indexed methods on benchmark instances from the literature. We also incorporate additional inequalities to guide the exploration of the branch-and-bound tree, yielding notable reductions in solution times and the number of explored nodes. Beyond minimal covers, we extend the clique-forbidding separation scheme in two directions: (a) inequalities derived from the extended covers of the same knapsack set, which requires addressing a Turán-type extremal problem for interval graphs, and (b) clique-forbidding constraints induced by arbitrary valid inequality of the knapsack polytope. Lastly, we propose a problem-specific cut selection scheme for these clique-forbidding constraints.

The proposed methods are extensively tested across diverse instance features, including scheduling horizons and problem dimensions, machine power distributions, peak-power thresholds, and ToU profile types. Among these, ToU profiles have the strongest impact, affecting both algorithmic performance and the behavior of the period-indexed formulation. Overall, the algorithmic framework performs consistently across both existing and new instances, solving many to optimality. On the instances not solved to optimality within the time limit, it achieves an average optimality gap of 1.6% and 5% on the most challenging subset. Additional analysis indicates that the formulation, together with the proposed valid inequalities, yields strong dual bounds.

The complexity introduced by specific ToU profiles motivates several directions for further research. To the best of our knowledge, this aspect has received limited attention in the literature, particularly in the context of period-indexed formulations. While the profiles considered mainly differ in the relative lengths of their time windows and exhibit a comparable number of periods, attributing performance differences to specific structural features of the tariff structures is not straightforward and calls for a more detailed combinatorial analysis. In parallel, the observed strength of the root relaxation suggests that further performance gains across ToU profiles are more likely to come from improvements on the primal side, motivating dedicated primal heuristics tailored to the proposed formulation and its lazy-cut integration. More broadly, future work could focus on establishing dominance relationships for the proposed valid inequalities and analyzing them from a polyhedral perspective. Further research could explore the extension of the problem-specific selection measures to these inequalities. The cut selection scheme could also be further tuned and studied to understand how cut-selection principles translate to settings where cuts are primarily used to restore feasibility rather than to strengthen dual bounds. The proposed approach could also be integrated into matheuristics to address larger-scale instances. Finally, because the separation scheme is largely independent of the underlying shop setting, the proposed inequalities and separation routines can be adapted to other machine and shop scheduling contexts. Consequently, this study lays the groundwork for the exact solution of a broader class of energy-constrained scheduling problems under ToU pricing.

CRediT authorship contribution statement

Marouane Felloussi: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Conceptualization. **Xavier Delorme:** Writing – review & editing, Supervision. **Paolo Gianessi:** Writing – original draft, Validation.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.ejor.2026.04.018.

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